

**FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING ( 25CS1201E)**
**2025-26, Term-III**
**Project Problem Statement Submission Form**

|                         |                 |                                                          |
|-------------------------|-----------------|----------------------------------------------------------|
| <b>Sec No-</b>          | <b>Team No-</b> | <b>Project Title- Hotel Revenue Management Dashboard</b> |
| Team Lead Roll Number   |                 | 2520030012                                               |
| Team Lead Name          |                 | DHARMIKA                                                 |
| Team Member Roll Number |                 | 2520030077                                               |
| Team Member Name        |                 | BINDHUMADHAVI                                            |
| Team Member Roll Number |                 | 2520030117                                               |
| Team Member Name        |                 | VAISHALI                                                 |

## 1. Abstract

The Hotel Revenue Management Dashboard is a visual analytics tool designed to centralize key metrics like ADR (Average Daily Rate), RevPAR (Revenue Per Available Room), and occupancy rates. Since hotel pricing and bookings fluctuate daily, manual tracking is difficult, time-consuming, and error-prone. This project addresses that problem by using charts and graphs to make complex data easy to understand at a glance. The system includes a Forecast Display to compare expected versus actual performance, and a Channel Breakdown to track booking sources. It also provides dynamic pricing suggestions based on demand levels and allows users to filter data and export reports. An admin panel enables data management and report generation. The project is organized into four modules: (1) Hotel Dashboard and KPI Monitoring, which displays ADR, RevPAR, occupancy rate, and booking pace in real time; (2) Revenue Trend Analysis and Forecasting, which uses graphical reports and historical data to support financial planning; (3) Booking Channel and Data Integration, which manages data from multiple booking platforms with real-time API synchronization; and (4) Pricing Strategy and Administrative Control, which handles dynamic pricing and user management. By providing a centralized, data-driven platform, the dashboard helps hotel managers make faster and smarter decisions to increase revenue.

## 2. Module Mapping

(Provide details of how modules are divided among team members.)

| <b>Module Name</b>                        | <b>Assigned Roll Number(s)</b>     | <b>Signature</b> |
|-------------------------------------------|------------------------------------|------------------|
| Hotel Dashboard & KPI Monitoring          | 2520030012                         |                  |
| Revenue Trend Analysis & Forecasting      | 2520030077                         |                  |
| Booking Channel & Data Integration        | 2520030117                         |                  |
| Pricing Strategy & Administrative Control | 2520030012, 2520030077, 2520030117 |                  |

### 3. Module Details

**Module 1 - Hotel Dashboard & KPI Monitoring:** Responsible for building the centralized dashboard interface that displays key performance indicators including ADR, RevPAR, occupancy rate, and booking pace through KPI cards and summary sections in real time.

**Module 2 - Revenue Trend Analysis & Forecasting:** Responsible for developing graphical reports and forecasting features that analyze revenue growth, occupancy trends, and future business predictions based on historical data to support strategic decision-making.

**Module 3 - Booking Channel & Data Integration:** Responsible for managing booking data collected from multiple channels such as online travel platforms and direct bookings, including channel-wise analysis, date-based filtering, and real-time API integration.

**Module 4 - Pricing Strategy & Administrative Control:** Jointly responsible for implementing dynamic pricing suggestions based on demand and occupancy levels, along with admin functionalities such as user management, report generation, system settings, and data export.

### 4. Frameworks / Platforms / Software Used

Frontend: HTML, CSS, JavaScript, React.js, Chart.js / Recharts (for data visualization). Backend: Node.js / Python (Flask or Django). Database: MySQL / MongoDB. API Integration: RESTful APIs for booking channel data synchronization. Tools & Platforms: VS Code, GitHub, Postman, Figma (UI design). Other: Excel / CSV for data import/export, PDF generation libraries for report export.

Guide Name and Signature (remarks if any) \_\_\_\_\_